HAMZA EL BYAD

MACHINE LEARNING ENGINEER / APPLIED AI / SCIENTIFIC COMPUTING

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PROFILE

Machine Learning Engineer with a PhD in Mathematics & Computer Science. Experience spanning industrial prediction, retrieval-augmented AI applications, statistical modeling, and scientific software. Combines research-oriented evaluation with practical Python engineering and CI/CD.

EXPERIENCE

Machine Learning Engineer | Rombo AI | 2025–2026

Cagliari, Italy — Industrial ML and NMR analytics; worked on a conversational product-catalog estimation assistant with retrieval, stateful workflows, deterministic cost logic and automated release testing.

Doctoral Researcher | University of Cagliari | 2022–2025

Italy — Scientific computing and statistical time-series methods applied to high-energy astrophysics; peer-reviewed publication output.

Data Science Research Intern | European Space Agency (ESAC) | 2023

Madrid, Spain — Applied deep-learning methods to compact-source detection in Planck sky maps.

Research Intern | INAF – Observatory of Cagliari | 2020–2021

Italy — Developed Python analysis for X-ray astronomical time series and timing signals.

SELECTED ENGINEERING WORK

CustomQuoteAI: RAG-powered catalog assistant with LangGraph, FastAPI, FAISS and Chainlit. Private v1.0.0 release, automated CI and documented browser acceptance.

NMR property prediction: spectroscopy preprocessing, PLS/PCR modeling and grouped validation. Reported experimental API-gravity result R^2 0.963, RMSE 1.312; no independent external validation.

SKILLS

ML: Python, PyTorch, scikit-learn, signal processing, regression, statistical validation. Applied AI: RAG, FAISS, LangGraph, FastAPI, Chainlit. Delivery: pytest, Git, GitHub Actions, Docker, TypeScript.

EDUCATION & RESEARCH

PhD, Mathematics & Computer Science — University of Cagliari. Selected first-author publication in The Astrophysical Journal (2025); co-authored scientific-computing research (2026).

Full case studies and selected publications in the accompanying portfolio.